IITM 2023T2

WEEK 2

PRACTICE QUESTIONS

Question 1

Write an SQL statement to find the name of the manager of the team: “All Stars”.

Database: FLIS

Solution:

```
select m.name
from managers m, teams t
where m.team_id = t.team_id and t.name = 'All Stars'
```

Question 2

Write an SQL statement to find the names of all teams.

Database: FLIS

Solution:

```
select name from teams
```

Question 3

Write an SQL statement to find the titles of books authored by an author having first name as ”Joh Paul” and last name as ”Mueller”.

Database: LIS

Solution:

```
select c.title
from book_catalogue c, book_authors a
where c.isbn_no = a.isbn_no
and author_fname = 'Joh Paul' and author_lname = 'Mueller'
```

Question 4

Write an SQL statement to find the titles of books published by "McGraw Hill Educa- tion".

Database: LIS

Solution:

```
select title
from book_catalogue
where publisher = 'McGraw Hill Education'
```

Question 5

Write an SQL statement to display the first name and the last name of students (stu- dent fname, student lname) pursuing 'PG' courses.

Database: LIS

Solution:

```
select student_fname, student_lname from students s, members m
where s.roll_no = m.roll_no and member_type = 'PG'
```

GRADED QUESTIONS

Question 1

Write an SQL statement to find the match numbers of those matches in which the host team scored more (goals) than the guest team.</br>
Database: FLIS

Solution:

```
select match_num
from matches
where host_team_score > guest_team_score;
```

Question 2

Write an SQL statement to find the colors of the home-jersey and the away-jersey (jersey home color, jersey away color) used by the team: “All Stars”.</br>
Database: FLIS

Solution:

```
select jersey_home_color, jersey_away_color
from teams
where name = 'All Stars';
```

Question 3

Write an SQL statement to find the names of players of the team: “All Stars”.</br>
Database: FLIS

Solution:

```
select p.name
from players p, teams t
where p.team_id = t.team_id and t.name = 'All Stars';
```

Question 4

Write an SQL statement to find the first names and the last names (student fname, student lname) of students who belong to the department with department code as
”MCA” and who were born after ’2002-06-15’.</br>
Database: LIS

Solution:

```
select student_fname, student_lname
from students
where department_code = ’MCA’ and dob> ’2002-06-15’;
```

Question 5

Write an SQL statement to find the first names and the last names of faculty (faculty fname, faculty lname) who belong to the department: ”Computer Science”.</br>
Database: LIS

Solution:

```
select f.faculty_fname, f.faculty_lname
from faculty f, departments d
where f.department_code = d.department_code
and department_name = ‘Computer Science’;
```

WEEK 3

PRACTICE QUESTIONS

Question 1

Write an SQL statement to find the names of teams that have played more than 3 matches.</br>

Database: FLIS

Solution:

```
select name from teams where team_id in
  (select team_id
  from ((select host_team_id as team_id, count(*) as c
  from matches
  group by host_team_id)
  union all
  (select guest_team_id as team_id, count(*) as c
  from matches
  group by guest_team_id)) as t1
  group by team_id having sum(c) > 3
  )
```

Question 2

Write an SQL statement to find the first name, last name of the faculty of the department
having department code as ”ME” and who have issued at least one book, such that there
are no duplicate firstname-lastname pairs.</br>
Database: LIS

Solution:

```
select distinct faculty_fname, faculty_lname
from faculty natural join members natural join book_issue
where faculty.department_code = ’ME’
```

Question 3

Write an SQL statement to find the number of book-titles issued on 11th August 2021.</br>
Database: LIS

Solution:

```
select count(title) from book_catalogue where isbn_no in
  (select isbn_no
  from book_issue a, book_copies b
  where a.accession_no=b.accession_no and
  a.doi='2021-08-11'
  )
```

Question 4

Write an SQL statement to find the names of faculty (faculty fname, faculty lname) who
did not issue any book.</br>
Database: LIS

Solution:

```
select faculty_fname,faculty_lname
from faculty
where faculty.id not in
(select members.id from members natural join book_issue
where members.id is not null)
```

Question 5

Write an SQL statement to find the unique book titles which are issued to “PG” students
but not to “UG” students.</br>
Database: LIS

Solution:

```
SELECT DISTINCT bc.title
FROM book_catalogue bc
JOIN book_copies bcp ON bc.ISBN_no = bcp.ISBN_no
JOIN book_issue bi ON bcp.accession_no = bi.accession_no
JOIN members m ON bi.member_no = m.member_no
JOIN students s ON m.roll_no = s.roll_no
WHERE s.degree LIKE 'M%'

EXCEPT
```

```
SELECT DISTINCT bc.title
FROM book_catalogue bc
JOIN book_copies bcp ON bc.ISBN_no = bcp.ISBN_no
JOIN book_issue bi ON bcp.accession_no = bi.accession_no
JOIN members m ON bi.member_no = m.member_no
JOIN students s ON m.roll_no = s.roll_no
WHERE s.degree LIKE 'B%';
```

GRADED QUESTIONS

Question 1

Write an SQL statement to find the names of teams that have played more than 3 matches.</br>
Database: FLIS

Solution:

```
select name from teams where team_id in
  (select team_id
  from ((select host_team_id as team_id, count(*) as c
  from matches
  group by host_team_id)
  union all
  (select guest_team_id as team_id, count(*) as c
  from matches
  group by guest_team_id)) as t1
  group by team_id having sum(c) > 3
  )
```

Question 2

Write an SQL statement to find the first name, last name of the faculty of the department
having department code as ”ME” and who have issued at least one book, such that there
are no duplicate firstname-lastname pairs.</br>
Database: LIS

Solution:

```
select distinct faculty_fname, faculty_lname
from faculty natural join members natural join book_issue
where faculty.department_code = 'ME'
```

Question 3

Write an SQL statement to find the number of book-titles issued on 11th August 2021.</br>
Database: LIS

Solution:

```
select count(title) from book_catalogue where isbn_no in
  (select isbn_no
  from book_issue a, book_copies b
  where a.accession_no=b.accession_no and
  a.doi='2021-08-11'
  )
```

Question 4

Write an SQL statement to find the names of faculty (faculty fname, faculty lname) who
did not issue any book.</br>
Database: LIS

Solution:

```
select faculty_fname,faculty_lname
from faculty
where faculty.id not in
(select members.id from members natural join book_issue
where members.id is not null)
```

Question 5

Write an SQL statement to find the unique book titles which are issued to “PG” students
but not to “UG” students.</br>
Database: LIS

Solution:

```
SELECT DISTINCT bc.title
FROM book_catalogue bc
JOIN book_copies bcp ON bc.ISBN_no = bcp.ISBN_no
JOIN book_issue bi ON bcp.accession_no = bi.accession_no
JOIN members m ON bi.member_no = m.member_no
JOIN students s ON m.roll_no = s.roll_no
WHERE s.degree LIKE 'M%'

EXCEPT

SELECT DISTINCT bc.title
FROM book_catalogue bc
JOIN book_copies bcp ON bc.ISBN_no = bcp.ISBN_no
JOIN book_issue bi ON bcp.accession_no = bi.accession_no
JOIN members m ON bi.member_no = m.member_no
JOIN students s ON m.roll_no = s.roll_no
WHERE s.degree LIKE 'B%';
```

WEEK 7

WEEK 7 OPPE Practice Question 1 - Prime Jersey

Problem Statement:

In this question, you have to write a Python program to print the names of the players and the team of each player of all those players whose jersey number is a prime number.

1. The list should be ordered in reverse alphabetical order of player names. If two or more players have the same name, then further sorting should be done on the team name, again in reverse alphabetical order.
2. The format of output is as given below:
   Name of the player, followed by a comma (,), then a space and then the team name.
   For example, if Arjun has jersey number 5 and is playing for All Stars and Pranav, with jersey number 7, is playing for team Amigos, then the output will be:
   Pranav, Amigos</br>
   Arjun, All Stars

Solution:

```
import sys
import os
import psycopg2

try:
   connection = psycopg2.connect(
       database = sys.argv[1],
       user = os.environ.get('PGUSER'),
       password = os.environ.get('PGPASSWORD'),
       host = os.environ.get('PGHOST'),
       port = os.environ.get('PGPORT'))

   cursor = connection.cursor()
   query = "select players.name, teams.name, players.jersey_n
o from teams, players where players.team_id = teams.team_id"

   cursor.execute(query)

   result = cursor.fetchall()

   def prime(n):
```

```
        if n<=1:
            return False
        for i in range(2,n):
            if n%i==0:
                return False
        return True
    final_res = []
    for res in result:
        if prime(res[2]):
            final_res.append(res[0]+", "+res[1])

    final_res.sort(reverse=True)

    for final in final_res:
        print(final)

    cursor.close()
except(Exception, psycopg2.DatabaseError) as error:
    print(error)
finally:
    connection.close
```

WEEK 7 OPPE Practice Question 2 - Encoding Teams

Problem Statement:

In this problem, you have to write a Python program to print an encoding of the ids of the teams whose jersey colour at home is different from the jersey colour when they play away from home.

- The encoding must be using a shift cipher, which is detailed below.

    - An alphabet is mapped to another alphabet as follows. For a given alphabet α, let pos be the position at which α occurs in the alphabet listing (A at 1, B at 2, ... Z at 26). Then the encoding of α is the alphabet at the position (pos + 7) mod 26.</br>
      For example, if M is the alphabet, then the position at which M occurs in the alphabet listing is 13. Then, the encoding of M is the alphabet at the position (13 + 7) mod 26 = 20, which is T.
    - For each digit β, the encoding of β is (β+7) mod 10.</br>
      For example, if 3 is the digit, then the encoding of 3 is the number (3 + 7) mod 10 = 0.
- The ids should be listed in the ascending order before performing the encoding.
- Each line in the output of the program must correspond to one row retrieved from the table.

Solution:

```
import sys
import os
import psycopg2

try:
   connection = psycopg2.connect(
       database = sys.argv[1],
       user = os.environ.get('PGUSER'),
       password = os.environ.get('PGPASSWORD'),
       host = os.environ.get('PGHOST'),
       port = os.environ.get('PGPORT'))

   cursor = connection.cursor()
   query = "select team_id from teams where jersey_home_color
<> jersey_away_color"

   cursor.execute(query)
```

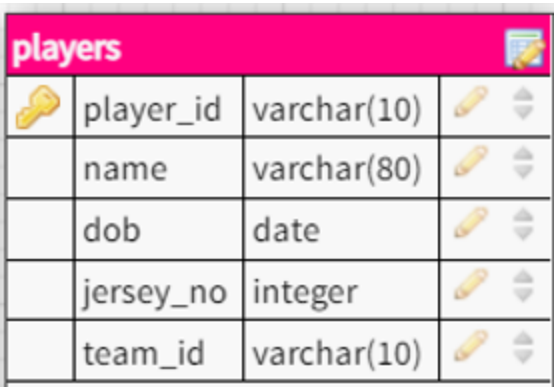
```
    result = cursor.fetchall()

    for res in result:
        new_res = chr(((ord(res[0][0])-65+7)%26)+65)
        for i in res[0][1:]:
            new_res += str((int(i)+7)%10)
        print(new_res)

    cursor.close()
except(Exception, psycopg2.DatabaseError) as error:
    print(error)
finally:
    connection.close
```

WEEK 7 OPPE Practice Question 3

Problem Statement:



players		
player_id	varchar(10)	
name	varchar(80)	
dob	date	
jersey_no	integer	
team_id	varchar(10)	

</br>
Write a Python program to output the jersey number of the player. Player's name is given in a file named 'player.txt' resides in the same folder as python program file. The output of the python program is only jersey number.

For example, if the jersey number of the player is 99. Then output must be 99 only. Note: No spaces.

Solution:

```
import sys
import os
import psycopg2

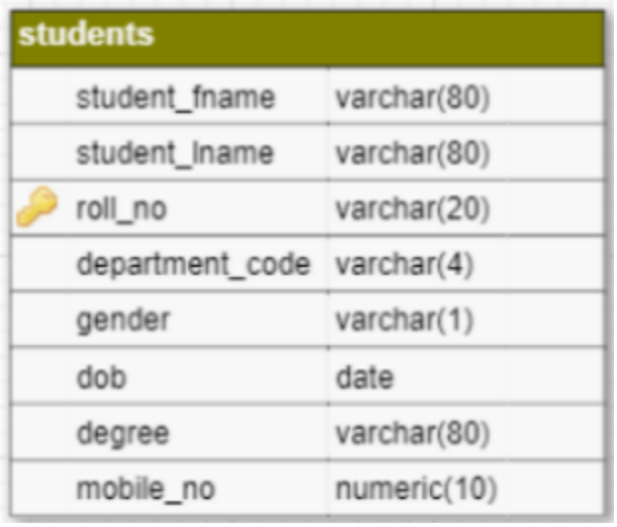
file  = open("player.txt" , "r")
name = file.read()

try:
   connection = psycopg2.connect(database = sys.argv[1],
   user = os.environ.get('PGUSER') ,
   password = os.environ.get('PGPASSWORD') ,
   host = os.environ.get('PGHOST'),
   port = os.environ.get('PGPORT'))

   cursor = connection.cursor()
   query = "select jersey_no from players where name = '{}'".
format(name)
   cursor.execute(query)
   result = cursor.fetchall()
   for res in result:
       print(res[0])
   cursor.close()
except(Exception, psycopg2.DatabaseError) as error:
   print(error)
finally:
   connection.close
```

WEEK 7 OPPE Practice Question 4

Problem Statement:



students	
student_fname	varchar(80)
student_lname	varchar(80)
roll_no	varchar(20)
department_code	varchar(4)
gender	varchar(1)
dob	date
degree	varchar(80)
mobile_no	numeric(10)

</br>
Write a Python program to print the roll number of the student. Student's first name is given in a file named 'name.txt' resides in the same folder as python program file.
The output of the python program is only roll number.

For example, if the first name of the student is 'Vikas'. Then output must be CS01 only. Note: No spaces.

Solution:

```
import sys
import os
import psycopg2

file = open("name.txt", "r")
name = file.read()
try:
   connection = psycopg2.connect(database = sys.argv[1],
   user = os.environ.get('PGUSER'),
   password = os.environ.get('PGPASSWORD'),
   host = os.environ.get('PGHOST'),
```

```
    port = os.environ.get('PGPORT'))

    cursor = connection.cursor()
    query="select roll_no from students where student_fname
='{}'".format(name)
    cursor.execute(query)
    result= cursor.fetchall()

    for res in result:
        print(res[0])
except(Exception,psycopg2.DatabaseError) as error:
    print(error)
finally:
    connection.close
```

WEEK 7 OPPE Practice Question 5

Problem Statement:



teams	
team_id	varchar(10)
name	varchar(80)
city	varchar(80)
playground	varchar(80)
jersey_home_color	varchar(80)
jersey_away_color	varchar(80)

</br>
Write a Python program to print the playground of the given team id. team_id is given in a file named 'team.txt' resides in the same folder as python program file.

- The output of the python program is only playground name.

- For example, if the team_id is 'T0002' . Then output must be Villa Park only

Solution:

```
import sys
import os
import psycopg2

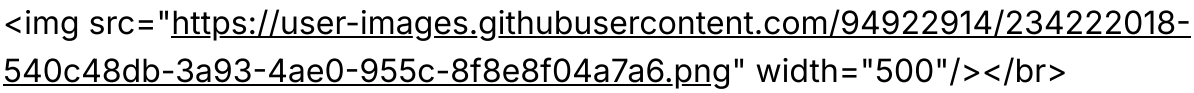
file = open("team.txt","r")
name=file.read()

try:
   connection = psycopg2.connect(database = sys.argv[1],
   user = os.environ.get('PGUSER') ,
   password = os.environ.get('PGPASSWORD') ,
   host = os.environ.get('PGHOST'),
   port = os.environ.get('PGPORT'))

   cursor=connection.cursor()
   query="select playground from teams where team_id='{}'".fo
rmat(name)
   cursor.execute(query)
   result=cursor.fetchall()
   for res in result:
       print(res[0])
   cursor.close()
except(Exception,psycopg2.DatabaseError) as error:
   print(error)
finally:
   connection.close
```

WEEK 7 OPPE Practice Question 6

Problem Statement:


Write a Python program to print the student's first name, the corresponding department name and the respective year of date of birth, if the year is even then print "Even" or else "Odd".
Student's first name is given in a file named 'name.txt' resides in the same folder as python program file.
The output of the python program is only student's first name, the corresponding department name and year of date of birth, if the year is even then "Even" or else "Odd".

- For example, 'Suman' and 'Computer Science' is the name and department name of the student. '2002' is the year he was born in. '2002' is even. Then, the final output will be Suman,Computer Science,Even only. Note: No spaces.
- For example, 'Vinod' and 'Electrical Engineering' is the name and department name of the student. '2003' is the year he was born in. '2003' is not even. Then, the final output will be Vinod,Electrical Engineering,Odd only. Note: No spaces.

Solution:

```
import sys
import os
import psycopg2

file = open("name.txt","r")
name = file.read()

try:
   connection = psycopg2.connect(
       database = sys.argv[1],
       user = os.environ.get('PGUSER'),
       password = os.environ.get('PGPASSWORD'),
       host = os.environ.get('PGHOST'),
```

```
        port = os.environ.get('PGPORT'))

    cursor = connection.cursor()
    query ="select students.student_fname, departments.departm
ent_name, students.dob from students,departments where studen
ts.student_fname = '{}' and departments.department_code = stu
dents.department_code".format(name)
    cursor.execute(query)

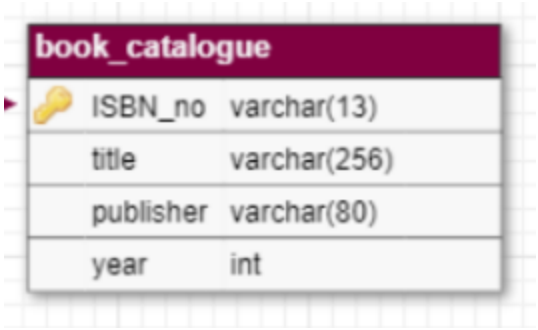
    result = cursor.fetchall()

    for res in result:
        if res[2].year%2==0:
            print(res[0]+","+res[1]+","+"Even")
        else:
            print(res[0]+","+res[1]+","+"Odd")

    cursor.close()
except(Execption, psycopg2.DatabaseError) as error:
    print(error)
finally:
    connection.close
```

OPE Python-PostgreSQL Question 1

Problem Statement:



</br>
Write a Python program to print the ISBN numbers of books which are published in a given year. Here, the year is obtained as the value of function L(x) (given after the sample output) at x. You have to read the value of x from the input file "number.txt", and use it to find the value of L(x). Your program must assume that the file number.txt resides in the same folder as your Python program.
You have to iterate through the list and print each value separately as shown in the output below:

- 9789352921171
- 9789351343202
- 9789353333380
  </br>
  The line function is given below:</br>
  L3(x) = 2000 + 5*x + 3

Solution:

```
import sys
import os
import psycopg2

file = open("number.txt","r")
x = file.read()

def L(x):
   return (2000+(5*x)+3)
```

```
year = L(int(x))

try:
   connection = psycopg2.connect(
       database = sys.argv[1],
       user = os.environ.get('PGUSER'),
       password = os.environ.get('PGPASSWORD'),
       host = os.environ.get('PGHOST'),
       port = os.environ.get('PGPORT'))

   cursor = connection.cursor()
   query = "select ISBN_no from book_catalogue where year =
'{}'".format(year)

   cursor.execute(query)

   result = cursor.fetchall()

   for res in result:
       print(res[0])

   cursor.close()
except(Exception, psycopg2.DatabaseError) as error:
   print(error)
finally:
   connection.close
```

OPE Python-PostgreSQL Question 2

Problem Statement:

Write a Python program to output the name of the main referee for a given match date (in yyyy-mm-dd format). The input to your program is a file named “date.txt”

that has the match date as the first word of the file. Your program must assume that date.txt resides in the same folder as your Python program.</br>
The output name has to be formatted as follows. The last name is displayed followed by the initials of the first name, then a full stop, a space and then the initials of the middle name (if the middle name exists), followed by a full stop.

- For example, if the name of the main referee is “Kennedy Sapam”, the output must be ”Sapam K.”
- If the name of the main referee is “Asit Kumar Sarkar”, the output must be ”Sarkar A. K.”
  </br>

Solution:

```
import sys
import os
import psycopg2

file = open('date.txt', 'r')
date = file.read()

try:
   connection = psycopg2.connect(
       database = sys.argv[1],
       user = os.environ.get('PGUSER'),
       password = os.environ.get('PGPASSWORD'),
       host = os.environ.get('PGHOST'),
       port = os.environ.get('PGPORT'))

   cursor = connection.cursor()

   query = "select referees.name from match_referees, matche
s, referees where match_referees.match_num = matches.match_nu
```

```
m and matches.match_date = '{}' and match_referees.referee =
referees.referee_id".format(date)

    cursor.execute(query)

    result_of_query = cursor.fetchall()

    for result in result_of_query:
        list_of_name = list(map(str, result[0].split(" ")))
        final_result = list_of_name[-1]
        for name in list_of_name[:-1]:
            final_result += " "+name[0]+"."
    print(final_result)

    cursor.close()
except(Exception, psycopg2.DatabaseError) as error:
    print(error)
finally:
    connection.close
```